Statistics

Chapter 9 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 10

Chapter 9 **Exercise 1** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **random sample** is a sample in which each item of the population had the same chance of being selected.

Result

Random sample

Chapter 9 Exercise 2

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

The **sampling error** is the error made when estimating a population parameter by a sample statistic. More precisely, the error is the difference between the population parameter and the sample statistic.

When considering the mean, then the difference between the population mean and the sample mean is thus called the **sampling error** as well.

Result

Sampling error

Chapter 9 Exercise 3

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

The standard deviation of the sampling distribution of the sample mean is called the **standard error**.

Result

Standard error

Chapter 9 Exercise 4

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

As the sample size increases, we have more information about the population and thus any estimates made about the population will be more accurate.

However, the variance of the sample means measures the accuracy of the sample means in estimating the population mean and thus the variance will **decrease** (as the estimates become more accurate).

Result

Become smaller

Chapter 9 Exercise 5

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **point estimate** is a single value that is used to estimate some population parameter.

For example, the sample mean is the point estimate of the population mean.

Result

Point estimate

Chapter 9 **Exercise 6** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **confidence interval** is a range of values in which we expect the true value of the population parameter to lie.

Result

Confidence interval

Chapter 9 Exercise 7

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

The width of the confidence interval is twice the margin of error.

The formula of the margin of error depends on the sample size, the variation in the population, and the level of confidence, while the formula of the margin of error does not depend on the population size (when not using the finite population correction).

This then implies that the width of the confidence interval is not affected by the **size of population**.

Result

Size of population

Chapter 9 Exercise 8

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **proportion** is a fraction of a population that has some particular property.

Result

Proportion

Chapter 9 **Exercise 9** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A t distribution is a continuous probability distribution, which has a mean of 0 (that is, the distribution is always centered at 0), and is dependent on the degree of freedom.

However, the t-distribution is not $\mathbf{positively\ skewed}$, because the t distribution is always symmetric (like the normal distribution).

Result

Positively skewed

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

When no preliminary estimate for the population proportion is available, then we use the population proportion 0.50 to determine the sample size because this population proportion will lead to the highest possible sample size (which is thus appropriate for all population proportions).

Result

0.50